

**What can we see
when we can see
overlap?**



Modeling Homer's *Iliad*
A MNML LMNL case study with exploratory markup

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Briareus design by John Flaxman (1793), engraved by Tommaso Piroli (1795)

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Briareus
Then call'd by thee, the Monster Titan came,
Published as the Act directs. Jan 22 1795, by J. Matthews Printer, Strand. (Whom Gods Briar-ars, Men Equan name.)

Pub. B. 1. 1. 522 Pope

ΡΑΨΩΔΙΑ Α

οὔνεκα τὸν Χρῦσιν ἠτίμασεν ἀρητῆρα
Ἀτρεΐδης· ὃ γὰρ ἦλθε θεὰς ἐπὶ νῆας Ἀχαιῶν
λυσόμενός τε θύγατρα φέρων τ' ἀπερείσι' ἄποινα,
στέμματ' ἔχων ἐν χερσὶν ἐκηβόλου Ἀπόλλωνος
χρυσέῳ ἀνὰ σκήπτρῳ, καὶ λίσσετο πάντας Ἀχαιοὺς,
Ἀτρεΐδα δὲ μάλιστα δύνω, κοσμήτορε λαῶν·
Ἀτρεΐδαι τε καὶ ἄλλοι εὐκνήμιδες Ἀχαιοί,
ὕμιν μὲν θεοὶ δοῖεν Ὀλύμπια δώματ' ἔχοντες
ἐκπέρσαι Πριάμοιο πόλιν, εὐ δ' οἴκαδ' ἰκέσθαι·
παῖδα δ' ἐμοὶ λύσαιτε φίλην, τὰ δ' ἄποινα δέχεσθαι,
ἄζόμενοι Διὸς υἱὸν ἐκηβόλον Ἀπόλλωνα.
ἐνθ' ἄλλοι μὲν πάντες ἐπευφήμησαν Ἀχαιοὶ
αἰδεῖσθαι θ' ἱερῇα καὶ ἀγλαὰ δέχθαι ἄποινα·
ἀλλ' οὐκ Ἀτρεΐδῃ Ἀγαμέμνονι ἦνδανε θυμῷ,
ἀλλὰ κακῶς ἀφίει, κρατερόν δ' ἐπὶ μῦθον ἔτελλε·
μή σε γέρον κοίλῃσιν ἐγὼ παρὰ νηυσὶ κιχείω
ἢ νῦν δηθύνοντ' ἢ ὕστερον αὐτὶς ἰόντα,
μή νύ τοι οὐ χραίσμη σκήπτρον καὶ στέμμα θεοῖο·
τὴν δ' ἐγὼ οὐ λύσω· πρὶν μιν καὶ γῆρας ἔπεισιν
ἡμετέρῳ ἐνὶ οἴκῳ ἐν Ἀργεῖ τηλόθι πάτρης
ἰστὸν ἐποιοχόμενῃν καὶ ἐμὸν λέχος ἀντιώωσαν·
ἀλλ' ἴθι μή μ' ἐρέθιζε σαώτερος ὥς κε νέηαι.
ὥς ἔφατ', ἔδεισεν δ' ὃ γέρον καὶ ἐπείθετο μύθῳ·
βῆ δ' ἀκέων παρὰ θῖνα πολυφλοίσβοιο θαλάσσης·

[BOOK [n]1{}}]

[para]{prayer [to]{Calliope}}{1 [n]1.1{}}μήνιν ἄειδε θεὰ Πηληϊάδεω Ἀχιλῆος{1}
[1 [n]1.2{}}ούλομένην, ἣ μυρὶ Ἀχαιοῖς ἄλγε' ἔθηκε,{1}
[1 [n]1.3{}}πολλὰς δ' ἰφθίμους ψυχὰς Ἄϊδι προΐαψεν{1}
[1 [n]1.4{}}ἡρώων, αὐτοὺς δὲ ἑλώρια τεῦχε κύνεσσιν{1}
[1 [n]1.5{}}οἰωνοῖσιν τε πᾶσι, Διὸς δ' ἐτελείετο βουλή,{1}
[1 [n]1.6{}}ἔξ οὗ δὴ τὰ πρῶτα διαστήτην ἐρίσαντε{1}
[1 [n]1.7{}}Ἀτρεΐδης τε ἀναξ ἀνδρῶν καὶ δῖος Ἀχιλλεύς,{1}{prayer}{para}
[para]{1 [n]1.8{}}τίς τ' ἄρ' σφωε θεῶν ἑριδοῖ ξυνέηκε μάχεσθαι,{1}
[1 [n]1.9{}}Λητοῦς καὶ Διὸς υἱός· ὁ γὰρ βασιλῆϊ χολωθεῖς{1}
[1 [n]1.10{}}νοῦσον ἀνὰ στρατὸν ὄρσε κακὴν, ὀλέκοντο δὲ λαοί,{1}
[1 [n]1.11{}}οὐνεκα τὸν Χρῦσην ἠτίμασεν Ἀργεΐῃρα{1}
[1 [n]1.12{}}Ἀτρεΐδης· [scene [set]{Achaian ships}{1}ὁ γὰρ ἦλθε θοὰς ἐπὶ νῆας Ἀχαιῶν{1}
[1 [n]1.13{}}λύσομένός τε θύγατρα φέρων τ' ἄπερείσι' ἄποινα,{1}
[1 [n]1.14{}}στέμματα' ἔχων ἐν χερσὶν ἐκηβόλου Ἀπόλλωνος{1}
[1 [n]1.15{}}χρυσῆν ἀνὰ σκήπτρῳ, καὶ λίσσετο πάντας Ἀχαιοὺς,{1}
[1 [n]1.16{}}Ἀτρεΐδα δὲ μάλιστα δῶν, κοσμήτορε λαῶν·{1}
[speech [by]{Chryses}[to]{Agamemnon, Menelaus, Achaians}{1}
[1 [n]1.17{}}Ἀτρεΐδαι τε καὶ ἄλλοι ἐϋκνήμιδες Ἀχαιοί,{1}
[1 [n]1.18{}}ὕμιν μὲν θεοὶ δοῖεν Ὀλύμπια δώματ' ἔχοντες{1}
[1 [n]1.19{}}ἐκπέρσαι Πριάμοιο πόλιν, εὖ δ' οἴκαδ' ἰκέσθαι·{1}
[1 [n]1.20{}}παῖδα δ' ἐμοὶ λύσαιτε φίλην, τὰ δ' ἄποινα δέχεσθαι,{1}
[1 [n]1.21{}}ἄζόμενοι Διὸς υἱὸν ἐκηβόλον Ἀπόλλωνα·{1}{speech}{para}
[para]{1 [n]1.22{}}ἐνθ' ἄλλοι μὲν πάντες ἐπευφήμησαν Ἀχαιοί{1}
[1 [n]1.23{}}αἰδεῖσθαι θ' ἱερῇα καὶ ἀγλαὰ δέχθαι ἄποινα·{1}
[1 [n]1.24{}}ἀλλ' οὐκ Ἀτρεΐδῃ Ἀγαμέμνονι ἦνδανε θυμῷ,{1}
[1 [n]1.25{}}ἀλλὰ κακῶς ἄφειλ, κρατερὸν δ' ἐπὶ μῦθον ἔτελλε·{1}
[speech [by]{Agamemnon}[to]{Chryses}{1}
[1 [n]1.26{}}μή σε γέρον κοίλῃσιν ἐγὼ παρὰ νηυσὶ κιχέω{1}
[1 [n]1.27{}}ἢ νῦν δηθύνοντ' ἢ ὕστερον αὖτις ἰόντα,{1}
[1 [n]1.28{}}μή νύ τοι οὐ χραίσμη σκήπτρον καὶ στέμμα θεοῖο·{1}
[1 [n]1.29{}}τὴν δ' ἐγὼ οὐ λύσω· πρὶν μιν καὶ γῆρας ἔπεισιν{1}
[1 [n]1.30{}}ἡμετέρῳ ἐνὶ οἴκῳ ἐν Ἀργεῖ τηλόθι πάτρης{1}
[1 [n]1.31{}}ἰστον ἐπιτοχομένην καὶ ἐμὸν λῆχος ἀντιώσωσαν·{1}
[1 [n]1.32{}}ἀλλ' ἴθι μὴ μ' ἐρέθιζε σῴτερος ὥς κε νέηαι·{1}{speech}{para}
[para]{1 [n]1.33{}}ὥς ἔφατ', ἔδεισεν δ' ὁ γέρον καὶ ἐπείθετο μῦθῳ·{1}{scene}
[scene [set]{seashore}{1}[1 [n]1.34{}}βῆ δ' Ἀχαιῶν παρὰ θῖνα πολυφλοίσβοιο θαλάσσης·{1}
[1 [n]1.35{}}πολλὰ δ' ἔπειτ' ἀπάνευθε κιὼν ἡρᾷθ' ὁ γεραιὸς{1}
[1 [n]1.36{}}Ἀπόλλωνι ἄνακτι, τὸν ἡὔκομος τέκε Λητώ·{1}
[prayer [to]{Apollo}[by]{Chryses}{1}
[1 [n]1.37{}}κλυθὶ μευ ἄργυρότοξ', ὃς Χρῦσιν ἀμφιβέβηκας{1}

What are we trying to achieve? Exploratory Markup

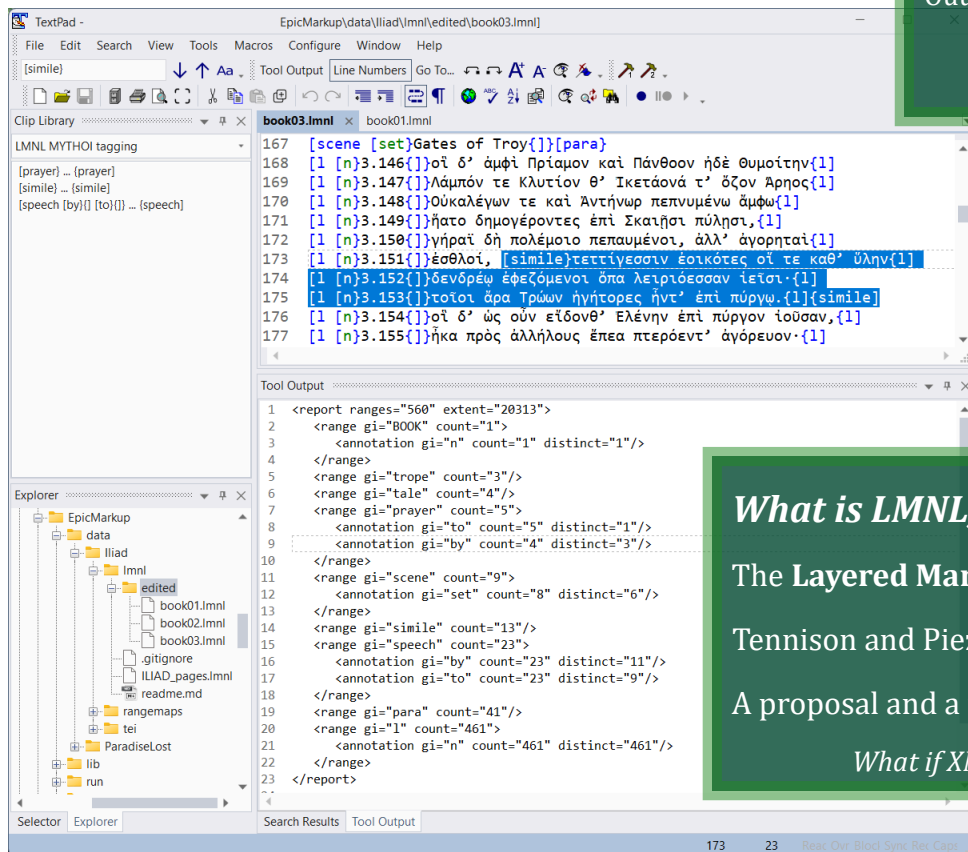
Unrealized potentials
of declarative text encoding
Describing data with tags
regarding overlap as normal
... We can tag freely as we go ...
A schema is an object of study
(not a binding commitment)
"More like guidelines"
Markup first, hierarchies later

Tools use **Laminator** XProc pipelines under XML Calabash or Morgana-IIIse (both Java)

But tagging is also fun and easy

Any plain-text (UTF-8) editor works well
For example, a **Bluefish** configuration is available
The tools also work on the command line
Or as an XProc application ...
... and a SaxonJS-based platform is contemplated

What if XML had no end-tag matching rule?



Laminator

<https://github.com/wendellpiez/Laminator>

How it works

An Invisible XML grammar plus handful of XSLTs

Glued together with XProc+iXML

(... only one way to do it ...)

LMNL syntax is read in via iXML parse plus XSLT x 2

XML can be ingested directly, or convert to LMNL

Results can be expressed in LMNL or XML syntax

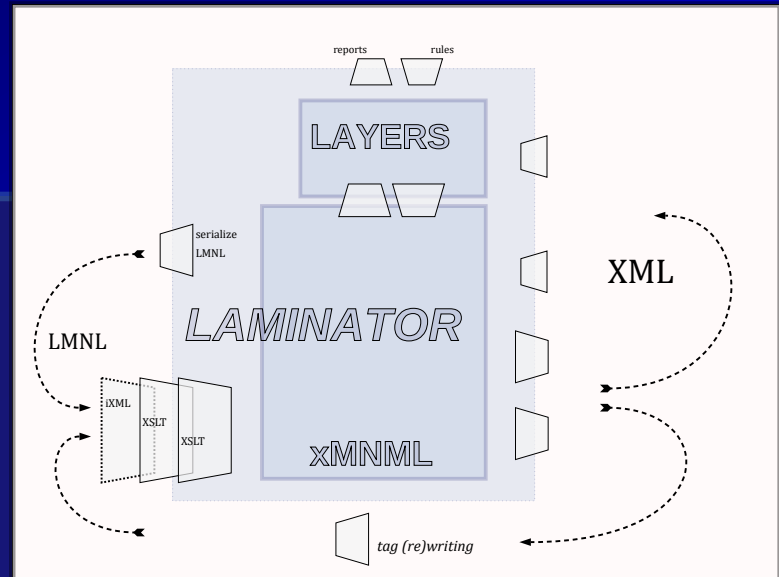
Works with any XML idiom for expressing overlap

Data flows are adaptable

problem domain

data set

tagging semantics



LMNL inputs

Made by hand

Or generated from (XML or other) source

[line [no]5.791{ }]ώς ἔφατ', [l1tt]οὐ δ' ἀπίθησε [nom]θεὰ[l1tt] γλαυκῶπις Ἀθήνη[nom].{line]

xMNML - a tagged text (sequence) model

```
PS C:\Users\wapie\Documents\Projects\Github\EpicMarkup\run> xc .\RUN-ONE-LINE_LMNL.xml
Invoking XML Calabash 3.0.54 for XProc 3 pipeline execution...
=====
xMNML :: 1 ::
<?xml version="1.0" encoding="UTF-8"?>
<LMNL xmlns="http://wendellpiez.com/ns/xMNML">
  <start gi="line" rID="r1_line" L="1" ch="1" off="0" ext="44">
    <annotation L="1" ch="7" gi="no">5.791</annotation>
  </start>
  <text cf="r1_line" off="0" ext="10">ὥς ἔφατ', </text>
  <start gi="l1tt" rID="r2_ltt" L="1" ch="20" off="18" ext="17"/>
  <text cf="r1_line r2_ltt" off="18" ext="10">οὐ δ' ἀπίθησε </text>
  <start gi="nom" rID="r3_nom" L="1" ch="48" off="24" ext="19"/>
  <text cf="r1_line r2_ltt r3_nom" off="24" ext="3">θεὰ</text>
  <end gi="l1tt" rID="r2_ltt" L="1" ch="56" off="27" ext="17"/>
  <text cf="r1_line r3_nom" off="27" ext="16">γλαυκῶπις Ἀθήνη</text>
  <end gi="nom" rID="r3_nom" L="1" ch="77" off="43" ext="19"/>
  <text cf="r1_line" off="43" ext="1">.</text>
  <end gi="line" rID="r1_line" L="1" ch="83" off="44" ext="44"/>
</LMNL>
=====
Elapsed time: 00:00:01.9967690
Press any key to continue...
```



LAYERS - a range model

```
PS C:\Users\wapie\Documents\Projects\Github\EpicMarkup\run> xc .\RUN-ONE-LINE_LMNL.xml
Invoking XML Calabash 3.0.54 for XProc 3 pipeline execution...
=====
LAYERS :: 1 ::
<?xml version="1.0" encoding="UTF-8"?>
<LAYERS xmlns="http://wendellpiez.com/ns/xMNML">
  <layer ranges="line ltt nom" fore="ech">
    <range rID="r1_line" gi="line" starting="0" ending="44">
      <annotation L="1" ch="7" gi="no">5.791</annotation>
    </range>
    <range rID="r2_ltt"
      gi="l1tt"
      starting="10"
      ending="27">
      overlaps="r3_nom"/>
    <range rID="r3_nom"
      gi="nom"
      starting="24"
      ending="43"
      overlaps="r2_ltt"/>
  </layer>
  <frontier>ὥς ἔφατ', οὐ δ' ἀπίθησε θεὰ γλαυκῶπις Ἀθήνη.</frontier>
</LAYERS>
=====
Elapsed time: 00:00:01.9888409
Press any key to continue...
```

XML results

... one (any) hierarchy at a time

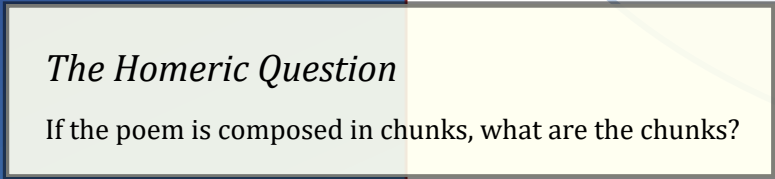
<line no="5.791">ώς ἔφατ', οὐ δ' ἀπίθησε <nom>θεὰ γλαυκῶπις Ἀθήνη</nom>.</line>

<line no="5.791">ώς ἔφατ', <l1tt>οὐ δ' ἀπίθησε θεὰ</l1tt> γλαυκῶπις Ἀθήνη.</line>

<line no="5.791">ώς ἔφατ', <l1tt>οὐ δ' ἀπίθησε <nom>θεὰ</nom></l1tt><nom> γλαυκῶπις Ἀθήνη</nom>.</line>

<line no="5.791">ώς ἔφατ', <l1tt_s>οὐ δ' ἀπίθησε <nom>θεὰ<l1tt_e/> γλαυκῶπις Ἀθήνη</nom>.</line>

Form and formula (*all the way down*) — WAVE or KYMA (κύμα) pattern



Range maps depict the extent of ranges of text within the poem.

Project goals and progress

<https://github.com/wendellpiez/EpicMarkup>

<https://github.com/wendellpiez/Laminator>

Free to inspect, assess, acquire, apply, adapt



Technical topics

Markup, tagged text models, range models and hierarchies

Whitespace (always)

Tag (re)writing vs/and transformation

Why XProc is nice for this

Parallel initiatives (by the developer)

Using public domain data from PerseusDL and Homer Multitext Project

Homer helper site (Scriptorium Method):

<https://raventracks.org/>

Lightweight *Iliad* Assisted Reader: **Iliad Elucidated**

https://raventracks.org/#elucidated_iliad

LMNL, MNML, Laminator

LMNL markup is now easy to make

From XML or from scratch

More data, more examples?

Applications: narrative and genre studies, textual criticism and others

TODO: Formal specifications for MNML LMNL as implemented

TODO: Laminator documentation and testing

TODO: Alternate implementations including XPath 4.0 (record objects), All-XSLT/iXML, no-XML?

TODO: LMNL Annotations; Hyperdocument model; XML Profiles; DTS 1.0 API ...?

To come: **Iliad Illuminated**

(Some day, by small steps)

Awaiting findings first

Ideas are welcome!

For now - range map diagrams and reading notes